

AI News Intelligence & Sentiment Analytics Dashboard Implementation Plan

This document provides a complete end-to-end implementation roadmap for building a real-time AI-powered news intelligence dashboard using NLP, sentiment analysis, and interactive visualization techniques.

1. Project Objective

Build a real-time dashboard that fetches live news articles based on user-entered topics and performs:

- Sentiment Analysis
 - News Trend Analysis
 - Keyword Extraction
 - Risk/Misinformation Scoring
 - Interactive Data Visualization
 - Real-time Dashboard Updates
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2. Final Features

Core Features:

- Topic-based live news search
 - Positive/Negative/Neutral sentiment classification
 - News Risk Meter
 - Word Cloud generation
 - Trending keyword analytics
 - News cards with source/date
 - Interactive charts and visualizations
 - Real-time updates via API
 - Deployable web application
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3. Recommended Tech Stack

Frontend: Streamlit

Backend: Python, Pandas

NLP: HuggingFace Transformers, NLTK, VADER

Visualization: Plotly, Matplotlib, WordCloud

API: NewsAPI or GNews API

Deployment: Streamlit Cloud / Render

Version Control: Git & GitHub

4. Folder Structure

```
AI-News-Dashboard/  
■■■ app.py  
■■■ requirements.txt  
■■■ README.md  
■■■ data/  
■■■ models/  
■■■ utils/  
■ ■■■ fetch_news.py  
■ ■■■ sentiment.py  
■ ■■■ preprocess.py  
■ ■■■ visualization.py  
■ ■■■ risk_score.py  
■■■ assets/  
■■■ screenshots/
```

5. APIs to Use

Preferred: NewsAPI

Website: <https://newsapi.org>

Alternative: GNews API

Website: <https://gnews.io>

Steps:

- Create free account
 - Generate API key
 - Store API key in environment variable or config file
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6. NLP Pipeline

Step 1: Fetch News Articles

Step 2: Clean text data

Step 3: Remove stopwords and symbols

Step 4: Run sentiment analysis

Step 5: Generate risk score

Step 6: Extract trending keywords

Step 7: Display analytics on dashboard

7. Sentiment Analysis Logic

Use pretrained NLP models for sentiment classification.

Output classes:

- Positive
- Neutral
- Negative

Suggested models:

- VADER Sentiment Analyzer

- HuggingFace transformers pipeline
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8. Risk Meter Logic

Build a simple custom scoring system:

- More negative headlines → higher risk
- Frequent alarming words increase score
- Calculate percentage of negative articles

Example:

Negative Sentiment > 70% → HIGH RISK

9. Dashboard Components

Main UI Components:

- Topic search box
 - Sentiment pie chart
 - Risk indicator card
 - Trending keywords section
 - Word cloud visualization
 - Latest news article cards
 - Interactive trend graphs
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10. Visualization Plan

Use Plotly for interactive charts:

- Pie chart for sentiment distribution
 - Bar chart for keyword frequencies
 - Line chart for sentiment trends
 - Gauge chart for risk meter
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11. Streamlit UI Flow

User enters topic → System fetches live news → NLP processing starts → Dashboard updates with:

- News cards
 - Charts
 - Sentiment analysis
 - Risk score
 - Insights
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12. Deployment Plan

Option 1: Streamlit Community Cloud

- Push project to GitHub
- Connect repository to Streamlit Cloud
- Deploy app

Option 2: Render

- Create web service
 - Add requirements.txt
 - Deploy application
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13. requirements.txt

```
streamlit
pandas
requests
plotly
transformers
nltk
wordcloud
matplotlib
scikit-learn
```

14. One-Day Execution Timeline

Phase 1 (2 Hours):

Setup project structure, APIs, dependencies

Phase 2 (2 Hours):

Build NLP pipeline and sentiment analysis

Phase 3 (2 Hours):

Create Streamlit dashboard and charts

Phase 4 (1 Hour):

Add styling, testing, and screenshots

Phase 5 (1 Hour):

GitHub upload + deployment

15. Expected Final Output

Deliverables:

- Fully working Streamlit web application
- GitHub repository
- Live deployed project link

- PPT presentation
 - Project report/documentation
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16. Resume Description

Developed a real-time AI-powered news intelligence dashboard that performs live sentiment analysis and misinformation-risk analytics using NLP, interactive visualizations, and live news APIs.

17. Future Improvements

- Add multilingual support
 - Add fake-news classification model
 - Integrate LLM summaries
 - Add historical sentiment tracking
 - Add user authentication
 - Add database support
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18. Project Comparison

Project	Difficulty	Real-Time	Resume Value	Demo Impact
Spam Detection	Easy	No	Medium	Medium
House Price Prediction	Easy	No	Medium	Low
News Intelligence Dashboard	Medium	Yes	High	High